Probability_2025

Question 01

Six unbiased coins are tossed.

(i) What is the probability of getting four heads?

Question 01

Six unbiased coins are tossed.

(ii) What is the probability of getting at least two heads?

Question 02

A fair die is rolled twice.

(i) What is the probability that the sum of the outcomes is 9?

Question 02

A fair die is rolled twice.

(ii) What is the probability that the sum of the outcomes is at least four?

Question 03

A card is drawn from a pack of cards. What is the probability that it is a:

(i) spade?

Question 03

A card is drawn from a pack of cards. What is the probability that it is a:

(ii) numbered card?

Question 03

A card is drawn from a pack of cards. What is the probability that it is a:

(iii) honour card?

Question 03

A card is drawn from a pack of cards. What is the probability that it is a:

(iv) spade or a honour card?

Question 04

If two cards are drawn from a pack of cards, what is the probability that they are

(i) from the same suit?

Question 04

If two cards are drawn from a pack of cards, what is the probability that they are

(ii) from different suits?

Question 05

All the numbered cards from a pack of cards are taken and shuffled. If two cards are drawn from them, what is the probability that they have the same colour but different numbers?

Question 06

A bag has fifteen balls of which six are red, five are black and remaining are green.

(i) If three balls are selected simultaneously, what is the probability that they are of different colours?

Question 06

A bag has fifteen balls of which six are red, five are black and remaining are green.

(ii) If three balls are selected successively (without replacement), what is the probability that they are of different colours?

Question 06

A bag has fifteen balls of which six are red, five are black and remaining are green.

(iii) If three balls are selected successively (with replacement), what is the probability that they are of different colours?

Question 07

Bag x contains five white balls and four blue balls. Another bag y contains four white balls and five blue balls.

(i) If one bag is selected at random and two balls are drawn from it, what is the probability that both are of the same colour?

Question 07

Bag x contains five white balls and four blue balls. Another bag y contains four white balls and five blue balls.

(ii) If one ball is selected from each bag, what is the probability that they are of different colours?

Question 08

If one number is selected from the first 100 natural numbers, what is the probability that it is divisible by 4 or 5?

Question 09

There are two factories I and II which produce 2000 and 5000 nuts respectively. Factory I and factory II have defect rates of 5% and 15% respectively. One defective nut is chosen at random from the set of defective nuts. Find the probability that it is from factory II.

Question 10

The odds in favour of A completing a task are 3 : 4, the odds against B completing the task are 2 : 5 and the odds against C not completing the task are 2 : 5.

(i) What is the probability that all of them complete the task?

Question 10

The odds in favour of A completing a task are 3 : 4, the odds against B completing the task are 2 : 5 and the odds against C not completing the task are 2 : 5.

(ii) What is the probability that none of them complete the task?

Question 10

The odds in favour of A completing a task are 3 : 4, the odds against B completing the task are 2 : 5 and the odds against C not completing the task are 2 : 5.

(iii) What is the probability that at least one of them will complete the task?

Question 10

The odds in favour of A completing a task are 3 : 4, the odds against B completing the task are 2 : 5 and the odds against C not completing the task are 2 : 5.

(iv) What is the probability that exactly one of them will complete the task?

Question 11

The probability that A can solve a problem is 3/5.

The probability that B can solve the same problem is 4/7. If both try to solve the problem, what is the probability that at least one of them solves the problem?

Question 12

There are six people in a room. What is the probability that exactly two of them were born on the same day of the week and everyone else was born on a different day?

Question 13

The probability of solving a question correctly is 2/5. Out of eight questions, find the probability that at least one question is solved correctly.

Question 14

A and B play a game of dice. Whoever rolls a composite number first wins the game. If A starts the game, what is the probability that B wins?

Question 15

(i) Raju rotates a roulette wheel which has markings from 1 to 20. If the wheel stops at a multiple of 3; he wins ₹15 and if it stops at a multiple of 7, he wins ₹20. Each time, Raju has to pay an amount of ₹4 as participation fee. In the long run, what is the average profit he makes per game?

Question 15

(ii) Ravi plays a game of dice. If an even number turns up, he is paid an amount of ₹15 and if an odd number turns up, he has to pay an amount of ₹25. Find the expected amount that he is paid/has to pay per game in the long run.

Question 16

A man walks only either one step forward or one step backward at a time. The probability of the man taking a step forward is 3/7 and probability of the man taking a step backward is 4/7.

(i) What is the probability that after 9 steps, he is one step away from his starting point?

Question 16

A man walks only either one step forward or one step backward at a time. The probability of the man taking a step forward is 3/7 and probability of the man taking a step backward is 4/7.

(ii) What is the probability that after 8 steps, he is one step away from his starting point?

Question 17

17. Four numbered cards are selected from a pack of cards. What is the probability that all the cards have the same number on them?

Question 18

18. Two small 1×1 squares are chosen at random from a 8×8 chessboard. What is the probability that they lie on the same diagonal?

Question 19

19. Bag A contains 6 red and 7 black balls. Bag B contains 5 red and 9 black balls. One ball is selected from A and put into B. Now one ball is selected from B. What is the probability that it is black?

Question 20

20. A test has five English questions and six Maths questions. A student attempts four questions. What is the probability that he attempted at least one Maths question?

Additional Questions

1. The ratio of the number of officers to workers in city A and city B is 7 : 3 and 4 : 9 respectively. One person is selected as a mayor of both cities A and B. The probability that he is from city A is 3/5. Find the probability that he is an officer.

Additional Questions

2. The probability of a bomb hitting a target is ¾. The least number of bombs that should be dropped to ensure that the probability of the target getting bombed is at least 0.95 is ______.

Additional Questions

3. Fifty coins are tossed. The probability of heads appearing on any of the coins is p. If the probability of heads appearing on 25 coins is same as that on 26 coins, find p.

Additional Questions

4. In a casino game involving a die, a player gets to roll the die 2000 times for an entry fee of ₹2,500. The player gets ₹4 each time he rolls an even number and has to pay ₹2 each time he rolls an odd number. If the die is biased and shows an even number four out of five times, should Rakesh play the game?

Additional Questions

5. Two men A and B decide to meet at a scheduled time between 4:00 p.m. and 5:00 p.m. It is decided that each will wait for the other for 10 minutes and if the other person doesn't come he would leave. What is the probability that the two men meet?